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Joining Ti-based metallic glass and crystalline titanium by magnetic pulse welding

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Highlights

- Ti was successfully joint with Ti-based metallic glass by magnetic pulse welding.
- Magnetic pulse welding preserves the disordered structure of metallic glass.
- Mixing zones at Ti/metallic glass interfaces contained Ti and Ti oxides and nitrides.
- The size of crystalline particles in mixing zones did not exceed 25 nm.

Abstract

Due to low thermal stability and limited critical size, metallic glasses (MGs) are frequently considered for reinforcing composites. The production technology of composites should provide the minimum heat input to preserve the disordered structure of MGs. In this study, the solid-state magnetic pulse welding (MPW) was used to join crystalline titanium and Ti-based MG. The amorphous structure of the MG layer after MPW was confirmed by synchrotron X-ray radiation diffraction (XRD), ultra-small-angle X-ray scattering (uSAXS), and transmission electron microscopy (TEM). Crystalline particles were found only in the mixing zones subjected to the strongest heating during welding. The average size of the crystalline precipitates was about 25 nm, and their phase composition corresponded to a-Ti. In addition to Ti particles, titanium oxides and nitrides could form at the interface of Ti and MG layers during MPW.

Keywords

Magnetic-pulse welding; High-velocity impact; Metallic glass composite; Synchrotron X-ray diffraction; Small angle X-ray scattering; Transmission electron microscopy

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1. Introduction

Metallic glasses (MGs) are known to have outstanding mechanical properties such as high tensile strength and fatigue resistance, enhanced impact strength, high resistance to wear and corrosion. The elastic strain limit of MGs is an order of magnitude higher than that of crystalline materials [1,2]. However, such alloys are characterized by a number of disadvantages, including low ductility and low thermal stability. In addition, MG production is a laborious task requiring high quenching rates necessary to retain a disordered structure [3]. This aspect limits the production of bulk MGs, and most often

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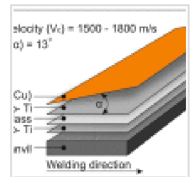
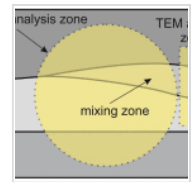
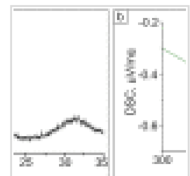
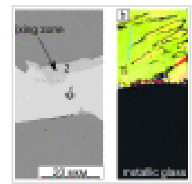
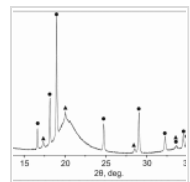
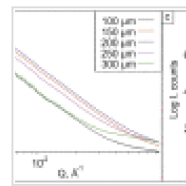

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Tables (2)

they are produced in the form of fine powders or thin ribbons. For this reason, MGs have found only limited application as structural materials.

One of the approaches that allows expanding the scope of MGs and using their advantages in load-bearing applications is the formation of MG-based composites [4,5,6,7]. However, fabrication of composites reinforced with amorphous components is also associated with various problems, first of them being the choice of an appropriate technological process for their formation. Up to date, a number of technologies allow obtaining amorphous/crystalline composites. Most notable are accumulative roll bonding [8], spark plasma sintering [9,10], hot pressing [11,12], and selective laser sintering [13]. The common drawback of these technologies is elevated processing temperatures that promotes crystallization of the glassy structure [14,15].

High-velocity impact welding, specifically explosion welding (EW) and magnetic pulse welding (MPW), can be considered as alternative methods for producing amorphous/crystalline composites [16,17,18,19]. Due to the short heating time and high cooling rates of joints, these methods do not force crystallization of the amorphous component. Despite the fact that EW and MPW are inherently related processes, the latter has a number of advantages, including the possibility of wide industrial applications, as this method does not require the use of explosives, and is generally cleaner and more precise [20].

Most of the studies on high-velocity impact welding are mainly focused on joining crystalline materials (such as Fe, Al, Cu, Ti, Ni, etc.) to each other [21], [22], [23], [24], [25], [26], [27], [28], [29]. A few works on bonding of MGs to crystalline materials by high-velocity impact were published in recent years showing high potential of such approach for the fabrication of amorphous/crystalline composites. For example, Zr-based bulk metal glass (BMG) with aluminum [30], Fe-based metallic glass with aluminum [31], BMG with copper [32,33] and some others combinations [34,35,36] were successfully joined by EW of MPW.

It can be noted that the vast majority of publications on high-velocity impact welding of glassy to crystalline alloys are devoted to EW. The structure of amorphous/crystalline composites obtained by MPW is rarely discussed in the literature. Currently, only a few studies have been published on the joining of Zr-based and Cu-based MGs with aluminum [37,38]. Thus, additional studies on the structure of amorphous/crystalline composites obtained by MPW will significantly supplement the existing information on the mechanisms of bond formation.

The purpose of this study was to obtain a Ti / Ti-based MG / Ti trimetal plates by MPW. The structure of the resulting composites was studied in detail using conventional metallographic methods as well as using synchrotron X-ray radiation.

2. Materials and methods

Commercially pure Ti foils (99.33 % Ti; 0.25 % Fe; 0.07 % C; 0.1 % Si; 0.04 % N; 0.2 % O; 0.01 % H, wt. %) with a thickness of 150 µm and a MG ribbon made of the STEMET® 1202 alloy (hereinafter referred to as 1202 alloy) with a composition of $Ti_{55.6}Ni_{10.5}Zr_{6.8}Cu_{17.8}Be_{8.6}V_{0.8}$ and a thickness of 50 µm [39] were used as starting materials. The MG foil was produced by melt spinning in the helium atmosphere using Kristall-702 apparatus described in detail elsewhere [40]. The melt was cooled on a copper disk rotating at a linear speed of 20 m/s at a cooling rate of 10^4 - 10^6 K/s.

The initial phase composition of MG was studied by X-ray diffraction (XRD) analysis using an ARL X'TRA laboratory diffractometer with a Mo X-ray tube. The diffraction pattern was recorded in a scanning mode with the 2θ step size of 0.02 ° and dwell time of 5 s per point. The crystallization temperature of MG was obtained by differential scanning calorimetry (DSC) with a SDTQ600 thermal analyzer. Heating was carried out in ceramic crucibles with an inert helium gas purge at a constant heating rate of 20 K/min. Amorphous foils were heated until the crystallization process was completed.

MPW was performed at the Lavrentiev Institute of Hydrodynamics, SB RAS (Novosibirsk, Russia) [16]. A capacitor battery with the capacitance of 3.4×10^{-3} F and voltage of up to 5 kV was used as a power source. The inductance of the battery and input cables was within 40 nH. Solid-state discharger initiated the battery discharge. The capacitor battery accumulated up to 40 kJ of energy. The discharge current in the experiment was recorded by an inductive sensor. During the experiment, the magnetic pressure, the driven plate velocity, and the distance covered by the plate were evaluated. These dependences made it possible to specify the impact velocity by choosing the distance between the driver plate and the upper foil in the welding assembly.

Welding was carried out according to the scheme shown in Fig. 1. To accelerate titanium and MG foils, a driver plate made of copper with a thickness of 0.5 mm and a width of 30 mm was used. The copper plate was not welded to the composite. The width of the MG and Ti foils was 20 mm. The length of the workpiece was 150 mm. The collision point velocity (V_c) and the collision angle between the copper plate and the upper Ti foil during the welding were 1500–1800 m/s and 13°, respectively.

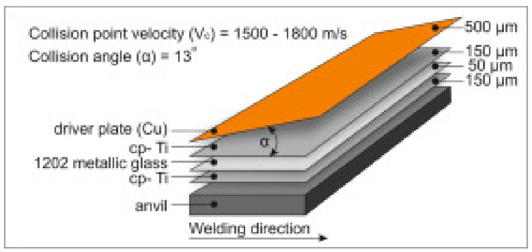

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Fig. 1. MPW scheme of the three-layer Ti / Ti-based MG / Ti composite.

The structure of the composite after the welding was studied by scanning electron microscopy (SEM) using a Carl Zeiss EVO 50 microscope coupled with an Oxford Instruments X-Act detector for energy-dispersive X-ray spectroscopy (EDX). Specimens for structural studies were prepared by grinding with abrasive papers (P120 – P2500) and alumina powder with a particle size of 5 and 1 μm followed by polishing using a colloidal silica. To study the structure of the crystalline part, the electron backscatter diffraction (EBSD) method was used. Investigations were carried out using a Carl Zeiss Sigma 300 SEM equipped with an Oxford Instruments HKL Channel 5 detector. The specimen for EBSD studies was prepared by grinding and polishing with abrasive papers, alumina powder, and a colloidal silica solution. At the final stage, the specimen was polished with argon ions using Technoorg Linda SEMPrep2 equipment.

The fine structure of the interface between Ti and the 1202 alloy was studied by transmission electron microscopy (TEM) using a Carl Zeiss Libra 120 microscope equipped with an Oxford Instruments X-Max 80T EDX detector. Specimens for TEM were prepared by the ion beam milling with gallium ions using a FEI Helios NanoLab 650 microscope. Zones subjected to analysis are shown in Fig. 2.

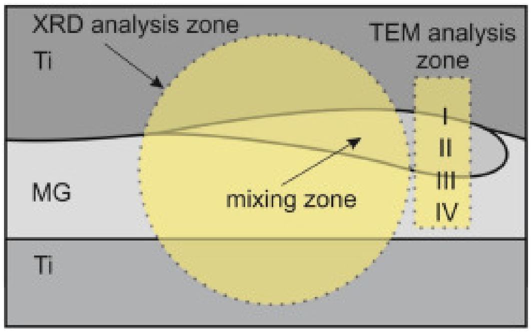

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Fig. 2. Schematical representation of the areas subjected to structural investigations. Numbers I-IV correspond to different zones of the composite and used as markers of these zone in TEM microphotographs and electron diffraction patterns.

The phase constitution of the sample after MPW was determined by synchrotron XRD. Measurements were carried out in transmission mode using a microbeam ($130 \times 130\,\mu m^2$) at the P03 beamline of the German Electron Synchrotron (DESY, Hamburg, Germany).

Before the measurements, the cross section of the sample was thinned to a thickness of 50 µm. The beam was centered on the 1202 layer, so that the entire width of the MG ribbon and partially titanium layers were in the area of analysis. The X-ray energy was 16.8 keV, which corresponded to a wavelength of 0.738 Å. Diffraction patterns were recorded by an DECTRIS EIGER 9M detector with a pixel size of $75\times75\,\mu m^2$. The sample-to-detector distance was 253 mm. Lanthanum hexaboride powder was used as a calibrant. Azimuthal integration was performed using PyFAI library for Python programming language [41]. The phase constitution and lattice parameters were determined using the PDF-4+ database and MAUD software [42].

Additionally, the material was studied by the ultra-small-angle X-ray scattering (uSAXS). The uSAXS data were recorded during linear scanning across the section of the sample with a step of 50 µm. In order to increase the statistical reliability, three line scans at a distance of 200 µm from each other were performed. A DECTRIS PILATUS 2M detector with the pixel size of $172\times172\,\mu m^2$ was used to record the scattering patterns. The X-ray energy was 6.79 keV which corresponded to the wavelength of 1.826 Å. The sample-to-detector distance was 4020 mm. Collagen was used for calibration. The patterns were azimuthally integrated using the GSAS-II software [43] and plotted in the “Log I – q^2” coordinates for further Guinier analysis.

The microhardness test was carried out using a 402 MVD (Wolpert Group) semi-automatic device. Measurements were performed across all three layers with a distance between indentations of about 30 µm. Microhardness of the mixing zones was measured separately. The load on the diamond pyramid was 10 g. Totally 5 measurements were carried out. The results are presented with a confidence interval.

3. Results

3.1. Characterization of the amorphous ribbon

The rapid solidification is known as one of the most effective ways to obtain amorphous metals [3]. There are several ways to choose the composition of a glass forming alloy. The most common approach is based on the following three empirical rules: 1) the alloy should consist of more than three elements; 2) difference in atomic size among the three main elements should be above 12 %; 3) mixing enthalpy among the three main constituent elements should be negative. The 1202 alloy meets these requirements. Thus, one can expect that the amorphous structure can be formed during rapid solidification of this alloy.

XRD pattern obtained from the ribbon after spinning had no diffraction peaks, but had a characteristic amorphous halo, which shows that MG possessed only short-range order (Fig. 3a). Consequently, one can conclude that the 1202 ribbon was in a completely amorphous state.

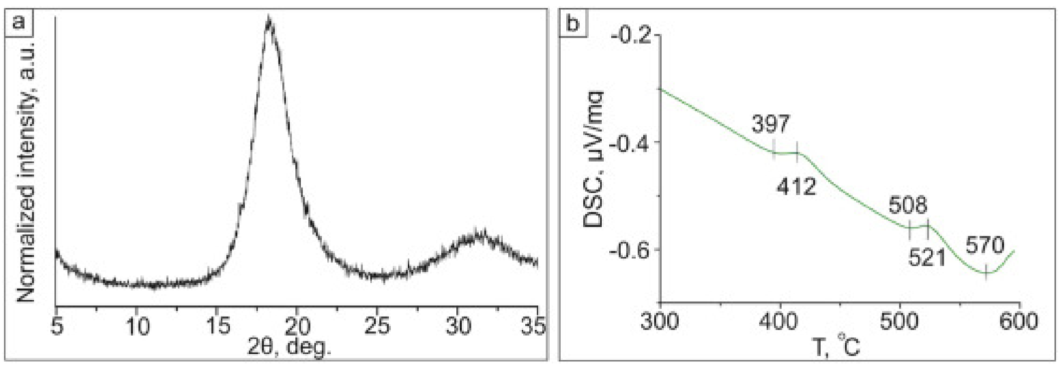

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Fig. 3. XRD pattern (a) and DSC curve (b) obtained from rapidly solidified 1202 alloy. The maximum intensity in the XRD pattern was equal to 320 counts /s.

Due to the metastable state, MGs are prone to crystallization at high temperatures [3, 13, 44]. For different compositions, the crystallization temperature (T_x) may vary. To determine the T_x for the 1202 alloy, thermal analysis was carried out. According to the results of DSC, crystallization of the 1202 alloy occurred in the temperature range of 397

– 570°C (Fig.3b). The glass transition temperature (T_g) for MG is usually 30 – 80°C lower than the temperature corresponding to the onset of crystallization. For Ti-based amorphous alloys, a value of T_g is 40 – 50°C lower than T_x [45,46]. Thus, to produce a composite with a completely amorphous MG component, it is necessary to avoid heating above 400°C, i.e., joining should be performed in a cold state. From this point of view, high-velocity impact welding is a reasonable approach for producing MG-based composites. This method causes heating only in narrow zones near the interface, and the temperatures exceeding the MG crystallization temperature are localized in small interfacial zones. At the same time, rapid heat transfer from these zones to the unheated material can cause vitrification of materials that are not even prone to amorphization in standard rapid solidification conditions. This is due to the fact that the major portion of the material remains unheated, which provides extremely high temperature gradients and, accordingly, high cooling rates [47]. These facts give grounds to believe that during MPW, the amorphous layer can be preserved in its original state, or, upon melting, it can undergo revitrification. The analysis of the three-layer composite provided in the next section confirms that such a hypothesis is possible.

3.2. Characterization of Ti / Ti-based MG / Ti composite after MPW

Under the dynamic impact during MPW, a three-layer composite consisting of two titanium layers and a 1202 alloy layer between them was formed. The structure of the composite in cross section along the direction of contact point movement is shown in Fig.4a. The structure of the upper and lower interfaces of the composite was slightly different. The lower boundary between titanium and 1202 alloy was flat. Along the upper interface the formation of mixing zones was observed, which is a typical phenomenon for high-velocity impact welding. The material in these zones often twists while in a liquid or plastic state, and cools rapidly due to heat transfer from these zones to the unheated surrounding material [48]. Twisting can be clearly seen in Fig.4a.

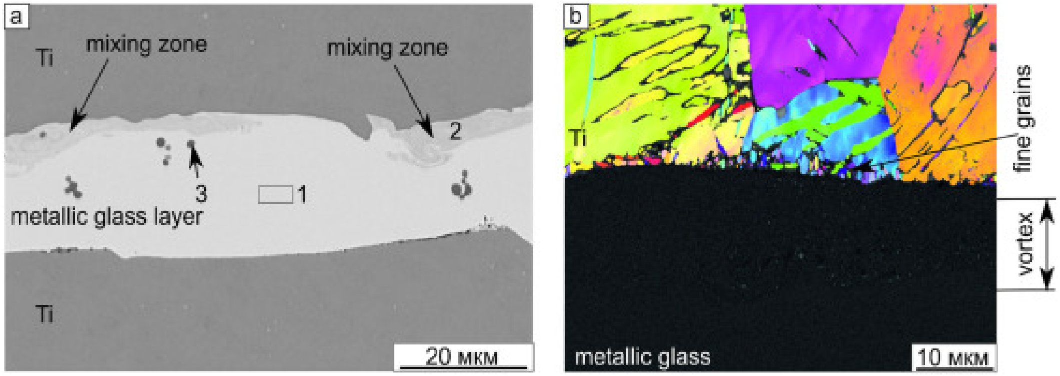

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Fig. 4. The structure of the Ti / Ti-based MG / Ti three-layer composite in the cross section (a) and the EBSD image (inverse pole figure map over band contrast map) (b).

EDX analysis showed that the composition of the middle layer approximately corresponded to the 1202 alloy (except for beryllium which is problematic for detection by this method) (zone 1 in Fig. 4a). The mixing zones contained a larger amount of titanium and smaller amounts of other alloying elements (zone 2 in Fig. 4a) compared with the 1202 layer, which confirms the mixing of two materials in these areas. The compositions of the zones marked with arrows are given in Table 1. In addition, large Ti-enriched precipitates were detected in the central part of the 1202 alloy layer. They could be formed at the stage of the amorphous ribbon producing (zone 3 in Fig. 4a).

Table 1. Elemental composition of the local areas of the Ti / Ti-based MG / Ti three-layer composite, at. %. Light elements (including Be) were neglected.

	Ti	Ni	Cu	Zr	V
Zone 1, Fig. 4a	59.3	11.6	19.0	9.2	0.9
Zone 2, Fig. 4a	67.8	8.9	15.1	7.5	0.7

	Ti	Ni	Cu	Zr	V
Zone 3, Fig.4a	91.4	1.7	2.8	4.1	-

Investigation of the upper Ti / 1202 interface containing a mixing zone revealed a number of structural features in the weld (Fig.4b). First, the formation of small titanium grains was observed near the interface. Their formation can be caused by one of the following factors. In the initial state, titanium had a coarse-grained structure. Upon impact, the welded plates were subjected to high-strain-rate deformation. The formation of twins in titanium grains are evidence of deformation processes. Near the interface the strains reached the highest values, and the heat released during the deformation promoted recrystallization in these layers [48]. However, under conditions of high-velocity impact welding, the primary crystallization is also possible. If the melting point of titanium was just reached in these local zones, then, upon cooling, the mixture of titanium grains surrounded with the 1202 alloy amorphous interlayers could form, which is indirectly evidenced by the dark contrast at the grain boundaries. A major part of the sample in the Fig.4a is dark. This indicates the absence of Kikuchi patterns, which is characteristic of amorphous materials. At the same time, small bright areas (i.e., small crystals) can been observed in the mixing zones. Thus, it can be concluded that a partially amorphous structure was formed there. However, the presence of an amorphous phase in the sample can be reliably identified only by XRD or TEM analysis.

Fig.5 shows an XRD pattern obtained from the cross section of the composite. The presence of a halo indicates the preservation of the amorphous structure in the 1202 layer. In addition to the halo, the XRD pattern contains a-Ti reflections obtained from titanium foils and some additional peaks. XRD analysis showed that these peaks corresponded to a compound with face-centered cubic (fcc) lattice ($Fm\bar{3}m$) with a lattice parameter equal to 4.224 Å. Since the amorphous ribbon consists of a large number of elements, it was not possible to unambiguously identify this phase. Zr-Cu and Zr-V compounds, as well as titanium oxides and nitrides, have this crystal structure and a similar lattice parameter. The appearance of additional peaks, as well as the EBSD data, may indicate partial crystallization of the amorphous ribbon.

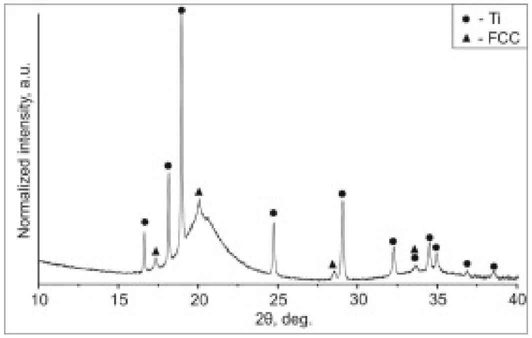

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Fig. 5. XRD pattern obtained from the composite cross section after MPW. The maximum intensity was equal to 50000 counts.

To analyze the size and shape of the precipitated particles, as well as their approximate composition, the uSAXS method in combination with TEM were applied. The uSAXS assumes the presence of inhomogeneities with a size of 1 – 100 nm in the investigated sample. Such inhomogeneities are responsible for X-ray scattering at small angles. Thus, the sheer fact that small-angle scattering occurs indicates the presence of small particles in the sample. In order to establish the fact of scattering, the integral of the obtained experimental signal was considered. The calculated values of the integral intensity from different zones of the sample are presented as a map in Fig.6a. The map shows that the integral intensity in the central part of the sample significantly exceeds that of the neighboring zones. Since the central part of the sample corresponds to the MG layer, one can conclude that this layer contains scattering inhomogeneities of nanometer size. Differences in the intensity of scattered radiation from the MG layer and neighboring zones corresponding to titanium are also clearly visible on the I vs. Q curves plotted in logarithmic coordinates (Fig.6b).

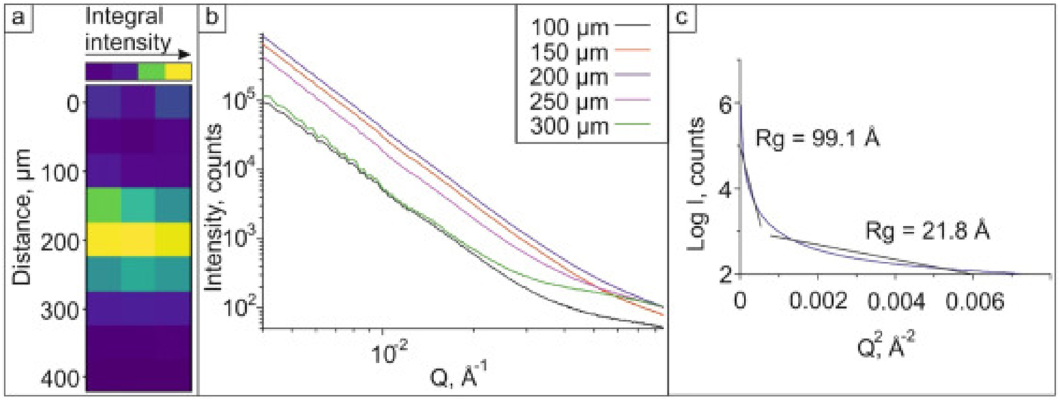

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Fig. 6. Results of the uSAXS investigations: a – change in the integral intensity in the cross section of the Ti / Ti-based MG / Ti composite; b – uSAXS curves obtained for different zones; c – Guinier plot for the 1202 layer.

To determine the size of the particles formed in the amorphous layer, scattering pattern from the 1202 zone was taken for further Guinier analysis. In this processing method, the size of the scattering inhomogeneities is reconstructed from the SAXS signal slope plotted in the Log I vs. q^2 coordinates in the Guinier approximation for spherical particles (Fig.6c):

$$I = I_0 exp\left(-\frac{(qR_g)^2}{3}\right) \tag{1}$$

where q is a scattering vector, and R_g is a radius of gyration, which is related to the particle size (D) as:

$$D = 2R_g\sqrt{\frac{5}{3}} \tag{2}$$

If the object under study has the complex structure and a non-monodisperse size distribution, the SAXS signal in the Log I – q^2 coordinates differs from a straight line, and approximation over the entire q range is able to provide information on some average particle size. The shape of the experimentally obtained scattering curve may indicate the presence of particles of two sizes. Approximation in the q range from 0.004 to 0.027 $Å^{-1}$ gave a gyration radius of 99.1 Å, which corresponds to a particle size of 25.5 nm. The approximation in the remaining range of q gave Rg=21.8 Å and D=5.6 nm.

In order to verify the data obtained by the uSAXS method, the fine structure of the samples was studied by TEM. The specimen for microscopy was prepared in such a way that a mixing zone appeared in the analysis area. Fig. 7a (zone I in Fig. 2) shows the interface between the 1202 alloy and titanium. The 1202 layer near the boundary zone consisted of many fine precipitates. At higher magnification, it can be seen that round-shaped particles have precipitated in the mixing zone (zone II in Fig. 2). The average size of these particles was about 23 nm (Fig. 7b). Thus, the size of the particles formed in the mixing zone correlated with the uSAXS data (25.5 nm) obtained in the q range of up to 0.027 $Å^{-1}$. The electron diffraction pattern obtained from the area shown in Fig. 7b indicated the formation of titanium particles in the mixing zone (Fig. 7c). It should be noted that the diffraction patterns also contained a halo, which indicates the presence of an amorphous matrix in the mixing zones.

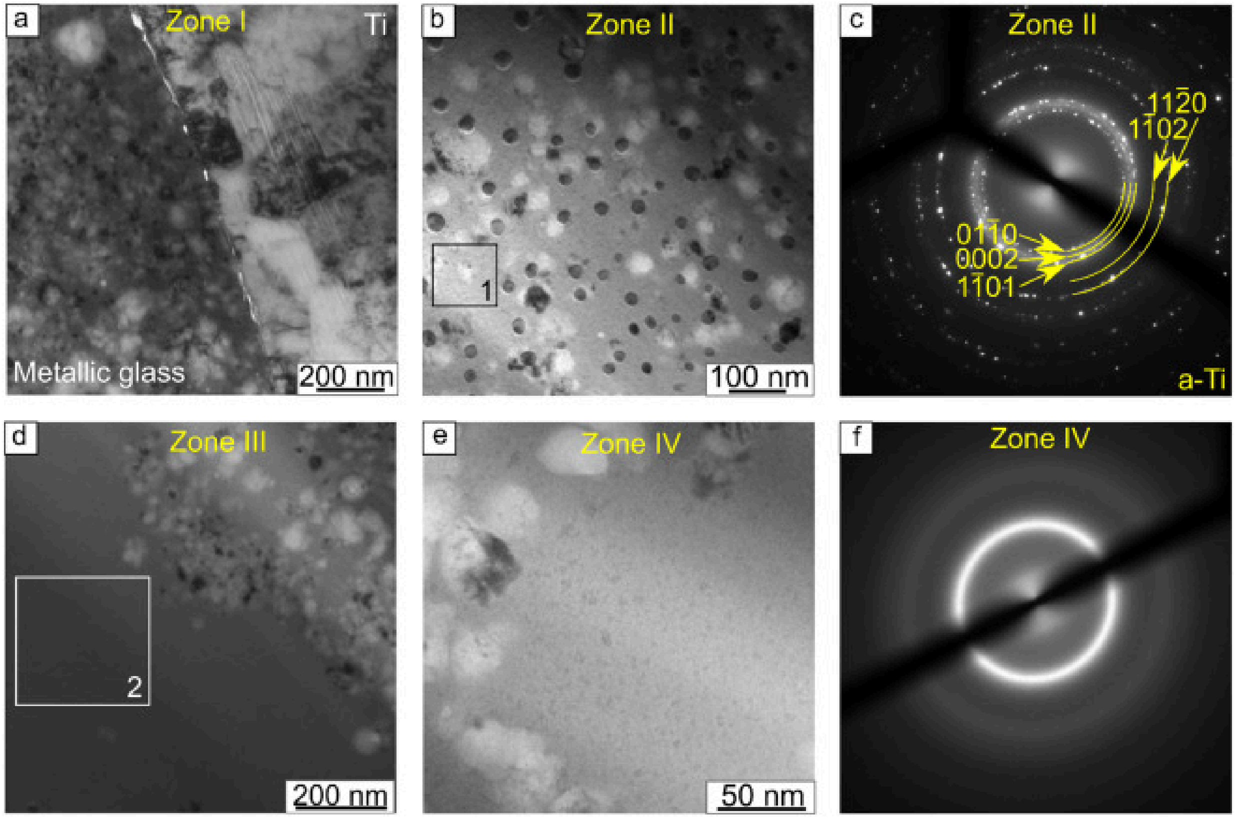

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Fig. 7. Results of the Ti / Ti-based MG / Ti composite investigations using TEM: a – an interface between Ti and 1202 alloy; b – nanoparticles precipitated in the mixing zone; c – the electron diffraction pattern obtained from the area shown in b; d – the interface between the mixing zone and the amorphous part of the 1202 layer; e – amorphous part of the 1202 layer; f – electron diffraction pattern obtained from the area shown in e.

In the direction from the interface to the center of the 1202 layer, a predominantly amorphous structure was observed which is confirmed by the electron diffraction pattern shown in Fig.7f, d clearly shows the boundary between the mixing zone and the amorphous part of the 1202 layer (zone III in Fig.2). Fig.7e shows the amorphous part of the 1202 layer at higher magnification (zone IV in Fig.2). However, at high magnifications, inhomogeneities become noticeable in the amorphous layer. Most probably, these inhomogeneities are nucleation centers of crystalline precipitates. The size of the precipitates did not exceed 5 nm, which is also in agreement with the uSAXS results.

At the same time, the origin and the composition of the fcc phase found in the structure of the composite by XRD (see Fig.5) remained unclear. An assumption can be made based on the EDX analysis of local areas marked on TEM micrographs (Fig.7). The results of the analysis are shown in Table2. The average composition of the 1202 alloy after welding with titanium was close to the initial composition of the alloy after melt spinning. At the same time, a high concentration of non-metallic elements, including nitrogen and oxygen, was observed in the layer. The maximum amount of nitrogen (29 at. %) was found in the mixing zone (zone 1 in Fig.7b). The major element in the composition of these zones is titanium, which suggests the presence of titanium nitride (TiN) in them. A high amount of oxygen may also indicate the formation of titanium oxide (TiO). However, the formation of nitrides in the mixing zones seems more probable, since oxygen was also found in zone 2 (Fig.7d) corresponding to the amorphous alloy in the initial state, while the remelted zones were saturated predominantly with nitrogen.

Table 2. Elemental composition of the local areas near the Ti – 1202 interface, at. %. The analyzed zones are shown in Fig.7.

	C	N	O	Al	Si	Ti	Ni	Cu	Zr
An average composition of the 1202 layer (including mixing zone)	9.7	6.8	16.4	0.3	0.4	40.5	4.3	17.8	3.8
Zone 1, Fig.7b	22.4	29	10.5	-	-	33.9	-	1.6	2.6
Zone 2, Fig.7d	19.2	-	12.3	0.3	0.4	39.5	7.2	15.3	5.8

3.3. Microhardness measurement results

Characterization of the composite revealed that the processes occurring at the interfaces of the welded materials during MPW significantly influenced the structure of both Ti and MG. In turn, these structural changes can affect the properties of the local areas of the composite. Due to the deformation, work-hardening of Ti during dynamic impact of plates during MPW may occur. Mixing zones can also contribute to the properties.

The microhardness distribution in the cross-section of the composite is shown in Fig. 8. An average microhardness of Ti layers is 195 HV. The trend towards increase in the microhardness of Ti near the interfaces with the MG foil is clearly seen. This phenomenon is explained by the deformation and corresponding work-hardening of the surface layers of plates due to high-velocity impact. Microhardness increase was also observed in the MG layer. An average microhardness of the MG foil was about 600 HV. Microhardness of the mixing zones was slightly higher (about 670 HV). As it was mentioned above, these zones experienced a partial crystallization and saturation with atmospheric gases. Both oxides and nitrides possess high hardness and could contribute to the reinforcement of the mixing zones. However, their influence was insignificant, while they were distributed in the amorphous matrix as separate nanoparticles. For this reason, one can conclude that the presence of the O- and N-containing compounds in small areas of the composite cannot significantly influence the properties of the material.

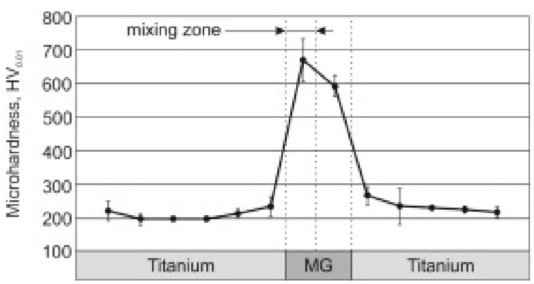

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Fig. 8. Microhardness distribution in the cross-section of the composite.

4. Discussion

A similarity in structure formation between dissimilar materials joined by MPW and EW was proven in multiple studies. In particular, the formation of waves and mixing zones, amorphization, and the appearance of metastable phases in the mixing zones, as well as the precipitation of nanosized particles, are the common features of these two processes. These phenomena were mentioned in a number of works on MPW [47,49,50] and EW [51,52,53] of dissimilar materials. It is explained by the fact that both processes are based on the same principle, namely high-velocity collision [20]. Under these conditions, a jet is formed between the colliding workpieces that cleans the surfaces from oxides and contaminants. At the same time, high pressures during the collision induce deformation of the metals for a short period of time. During welding, two metal surfaces cleaned by the jet of its oxide layers are made to come in close contact with each other due to the application of very high pressures and form interatomic bonding. Moreover, in both EW and MPW, the same process parameters influence the resultant bond. Thus, due to the insufficient number of studies on MPW of crystalline alloys with MGs, it is reasonable to take into account the literature data on EW when analyzing the results obtained.

According to our data, clearly distinguishable waves did not form during MPW of an amorphous ribbon with crystalline foils. This is probably due to the high rigidity and strength of the amorphous ribbon, which is hardly prone to plastic flow [12, 46]. At the same time, the loading during welding was excessive, since the formation of rather large mixing zones was observed at the upper interface. During the welding of the Al alloy with BMG, Hanliang et al. [54] controlled the thickness of the explosive, thus changing the collision velocity and the collision angle of the plates. With an increase in these parameters, the amount of kinetic energy dissipated during the collision (energy input) rises. Consequently, with increasing kinetic energy loss, the size of the mixing zones grows. At the same time, welding of multilayer composites has its own particularities. Specifically, in the direction from the explosive initiation plane to the lower layers, the amount of dissipated kinetic energy decreases [55], which explains a flat lower interface between the Ti and 1202 layers.

Mixing zones are subjected to intense heating. Heat generated by plastic deformation near the collision point leads to a temperature rise. Depending on collision regimes the temperature can be as high as melting point of welded plates. According to estimation made by Geng et. al [47], the heating temperature during MPW of iron with aluminum reached 1200 K, while the simulation of the EW process gives much higher values [48]. Although the temperature at the interface definitely exceeds the crystallization temperature of the 1202 amorphous alloy, its complete devitrification did not occur even in the mixing zones, which is associated with the high cooling rates. The cooling rates in the mixing zones significantly depend on their size and the thermophysical properties of the materials being welded. The cooling rate of a mixing zone with the size of approximately 1-10 μm under EW conditions can reach 10^6 – 10^7 K/s, which is comparable to the cooling rate during the melt spinning [35, 48, 56]. In the study [47], authors estimated that the cooling rate can be as high as 10^{14} – 10^{15} K/s for mixing zones with a size of 10 nm. Such high cooling rates lead to the melting of materials during the welding process followed by revitrification (partial or complete) during cooling, and even when fully crystalline materials are welded, their vitrification can occur [47, 50].

XRD studies of the Ti / Ti-based MG / Ti composite carried out in the present work confirm this fact, since both the XRD patterns (Fig.5) and the electron diffraction patterns (Fig.7) show a halo from the amorphous phase. However, even though crystallization of the MG did not occur, titanium reflections were found in the mixing zones. Watanabe and Kumai [49] observed a similar effect examining Al and Ti welds. The mixing zones contained fine precipitates with a phase composition corresponding to aluminum. However, authors assumed that most probably supersaturated Al-Ti solid solutions could form in these conditions. Thus, the time of the mixing zone being in the heated (melted) state was so short that no reaction could develop between the components to be welded.

EW of amorphous and crystalline materials quite often proceeds according to this scenario. For example, at the welding of copper and BMG, copper nanoparticles precipitated in the mixing zone [32]. In the direction from the Cu-BMG interface to the central part of the mixing zone, the crystallinity degree (i.e. the amount of copper inclusions in the mixing zone) decreased. When cladding aluminum with MG, Liu et al. [35] also observed gradual decline in the degree of crystallinity in the MG layer in the direction from the boundary with Al. However, phase composition of the crystalline component was not mentioned in this study. On the other hand, properly selected welding regimes allow minimizing the size of mixing zones and achieving a flat interface without any crystalline particles in the MG layer [57].

Due to the aforementioned features of the process, nanocrystalline particles (5 nm, Fig. 7e) seem unlikely to form outside the mixing zone during MPW. Most probably, their formation occurred at the production stage of the amorphous ribbon. As the SAED method did not allow determining their phase composition, the central part of the amorphous layer requires a more detailed analysis using high-resolution TEM.

The formation of oxides and nitrides in mixing zones remains questionable. The oxygen- and nitrogen-based compounds in mixing zones are very rarely discussed in the literature. Yu et al. found that mixing zones produced by MPW of SS 304 steel and 5A02 aluminum alloy contain elevated amount of oxygen [50]. In view of the fact that the highest temperatures and local melting of the material are reached near the collision point, the molten material has the possibility of being directly in contact with the air atmosphere. Considering the extremely high reactivity of molten titanium, the formation of oxides and nitrides in the mixing zones can be expected.

Summarizing these factors, one can conclude that MG are quite reliably welded to titanium by MPW. However, to form a weld of a better quality, more careful adjustment of the process parameters is required.

5. Conclusions

The three-layer composites were successfully fabricated by magnetic pulse welding of Ti foils and MG ribbon made of 1202 alloy. At the upper interface subjected to more severe welding conditions, mixing zones were formed. These zones contained an increased amount of titanium compared to the initial composition of MG due to the mixing of welded materials. In a thin layer of titanium adjacent to the mixing zone, the fine-grained structure was observed. Both foils experienced high-strain-rate deformation during welding. In the case of Ti, it was evidenced by the formation of twins in the grains. The use of magnetic pulse welding made it possible to preserve the amorphous structure of the 1202 alloy. Nanosized crystalline precipitates were found only in the mixing zones. The size of the precipitates did not exceed 25 nm. Diffraction analysis identified these precipitates as a-Ti. In addition to a-Ti, fcc phase with a lattice parameter of 4.24 Å was found at the interface. Since the energy-dispersive analysis of these zones showed an increased amount of oxygen and nitrogen, it can be assumed that during welding, titanium was saturated with atmospheric gases, which in turn caused the formation of titanium oxides and nitrides.

CRediT authorship contribution statement

D.V. Lazurenko: Conceptualization, Formal analysis, Writing – original draft, Investigation, Supervision, Funding acquisition. **A.A. Ivannikov:** Conceptualization, Investigation, Data curation, Formal analysis, Writing – review & editing. **A.G. Anisimov:** Resources, Investigation, Writing – review & editing. **N.S. Popov:** Investigation, Validation, Visualization, Writing – original draft. **G.D. Dovzhenko:** Formal analysis, Validation. **I.A. Bataev:** Investigation, Formal analysis, Writing – review & editing. **K.I. Emurlaev:** Investigation, Validation. **T.S. Ogneva:** Formal analysis, Data curation. **E.D. Golovin:** Investigation, Methodology.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests, Daria Lazurenko reports financial support was provided by Russian Science Foundation

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Data Availability

Data will be made available on request.

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> …Nevertheless, in certain dissimilar welding cases, a remarkable increase in hardness at the joint interface has been observed. Examples of such cases include magnetic pulse welding of a Ti-based BMG to pure Ti foils [533], EBW of a Zr-based BMG to Ti [471], and explosive welding of a Ti-based BMG to

crystalline aluminum [535] and a Zr-based BMG to crystalline copper [540]. The increase in hardness at the joint interface is generally an indication of induced nanocrystallization or minute diffusion of elements from the crystalline PM to the weld-affected zone.…

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Zhou Y., ..., Wang X.

Journal of Materials Research and Technology • Volume 30 • 2024

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…Li analyzed the effects of the horizontal and vertical components of the impact velocity on the interface morphology and evolution during the EMPW process based on a hydrodynamic model [10]. Lazurenko et al. achieved three-layer composites fabricated by EMPW of Ti foils and MG ribbon made of 1202 alloy [11]. Bembalge et al. welded the Al and steel tubular components by the EMPW under discharge voltages of 12–14 kV and at an air gap of 1–2.5 mm [12].…

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Alloying strategies for additive manufacturing of Ti6Al4V based alloys, composites and functionally graded materials: Microstructure and phase evolution of intra and inter-layer

Shi Y., ..., Qi B.

Journal of Materials Research and Technology • Volume 27 • 2023

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…In addition to the addition of alloying elements, the second phase reinforcement can also improve the wear resistance, microhardness, strength, and service temperature of Ti6Al4V. Among them, particle reinforcement is the most commonly used, which can be divided into (i) ceramic particles such as oxides (Fe_3O_4、ZrO_2、TiO_2、Al_2O_3 and rare earth metal oxides: Y_2O_3、La_2O_3), carbides (WC、TiC、SiC、B_4C、Cr_3C_2), borides (TiB、TiB_2、CrB_2), nitrides (TiN、BN、Si_3N_4) and intermetallic compounds (TiAl、Ti_5Si_3) [109–111]; (ii) metallic glasses (BMG), such as Zr-based, Fe-based and Cu-based [112]; (iii) high entropy alloys (HEAs), consisting of refractory HEAs of transitional metals, high-entropy catalytic nano-particles and CoCrFeNi-based, respectively [42,113]. Table 2 lists the physical properties of reinforced particles that can form titanium matrix composites by non-mechanical mixing, mechanical mixing, and in-situ addition with reactive non-mixing into Ti6Al4V [54,56,111].…

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> …Welding technologies have effectively scaled up bulk metallic glasses (BMGs) [1–3]. For example, the welding methods of dissimilar materials like BMGs and crystalline alloys such as Cu, Ti, Al, and 316 L were successfully developed, including vaporizing foil actuator welding [4], friction stir welding [5–12], magnetic pulse welding [13,14], ultrasonic welding [15–17], pulse current welding [18,19], explosive welding [20,21], laser welding [22–24], electron beam welding [25] and liquid-solid joining [26]. The joining of similar materials between identical BMGs or different BMGs was also investigated by ultrasonic welding, laser welding, and pulse current welding.…

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